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Uncertainty of a Titration Equivalence Point

A Graphical Method Using Spreadsheets To Predict Values and Detect Systematic Errors

Lowell M. Schwartz
University of Massachusetts, Boston, MA 02125

One aspect of teaching pH potentiometric titration methodology in analytical chemistry has troubled us over the years: the difficulty of assigning an uncertainty estimate to an equivalence point determined from a titration curve. Early in the course we emphasize that error estimates must be done for all quantitative determinations in science. Then we proceed to show students how to calculate an endpoint from an inflection point or an equivalence point from a Gran plot, but we fail to offer a useful way of estimating the uncertainties. Nonlinear regression methods (*1*) are a well-known means of estimating titration parameter uncertainties, but these are much too computer-intensive to include when titration analyses are first introduced.

A New Approach to Estimating Uncertainties

In this paper we describe a procedure that can be used to estimate the uncertainty of the equivalence point of a titration using graphical displays rather than nonlinear regression. While the procedure can, in principle, be carried out with a hand calculator and graph paper, the work becomes tedious for titrations involving many titration points. The ideal tool is an electronic spreadsheet that can display graphs on the screen. While the example offered here is a simple acid–base titration, the procedure can be extended to titrations involving multiprotic analytes, mixed analytes, activity coefficients, and other complications.

A potentiometric titration curve consists of a series of points of pH or cell potential measurements vs. added volume of titrant. The method is based on the fact that if all other titration parameters are known, it is always possible to calculate a prediction of the equivalence-point (ep) volume from each titration point. If all potentiometric and titrant volume measurements are free of error and if the parameters are accurately known, all such predictions must yield the same value for the ep volume.

However, if the parameters are accurately known but the potentiometric and titrant volume measurements include random errors, these errors propagate to the predictions, yielding a set of ep volumes perturbed by random error. From these a statistical uncertainty can be calculated. If the parameters are not accurately known, in other words, if the parameters suffer from systematic (determinate) errors, these errors also propagate to the predicted ep volumes yielding a set of values that vary along the titration curve in a regular, as opposed to random, way.

As will be seen, the method offers a convenient graphical way of judging the presence or absence of these systematic errors. In fact, if one objective of the titration is to determine a value for some parameter other than the ep volume, the particular parameter value that propagates no systematic error to the set of ep volumes serves to determine that parameter value. The following equations are proposed to model the example, which is described below.

Model Equations

Suppose that an initial analyte solution of volume V_0 mL of a monoprotic acid HA is titrated with strong base, say NaOH, of concentration C M. The volume of added titrant is v mL, and when v^* mL are added, the acid is completely neutralized. Thus, the analyte solution contains v^*C mmol of acid.

The titration is done under conditions of constant ionic strength so that all activity coefficients are invariant with changes in concentration during the titration. In order for the equations to accomodate strong acids more conveniently, the acid dissociation equilibrium is viewed as the following association reaction.

$$H^+ + A^- \rightleftarrows HA$$

Its conditional stability constant K, expressed in terms of concentrations, is

$$K = \frac{[HA]}{[H^+][A^-]} \tag{1}$$

This K is the reciprocal of the conventional acid dissociation constant K_a. It is 0 for a rigorously strong acid.

The conditional autoprotolysis constant for water is K_w = $[H^+][OH^-]$. The pH meter is calibrated so that the pH readings measure the hydrogen ion concentration directly through the equation

$$pH = -\log [H^+]$$

or

$$[H^+] = 10^{-pH}$$

At any point during the titration the total volume of analyte solution is

$$V = V_0 + v$$

and the millimoles of HA and A^- total to

$$v^*C = ([A^-] + [HA])V \tag{2}$$

Assuming that the only ionic species present are A^-, OH^-, Na^+, and H^+, the charge balance is

$$[A^-] + [OH^-] = \frac{vC}{V} + [H^+] \tag{3}$$

We solve for v^* by substituting $[A^-]$ from eq 3 into eqs 1 and 2. Then we substitute [HA] from eq 1 into eq 2 to get the following result.

$$v^* = \left\{v + \left([H^+] - [OH^-]\right)\frac{V}{C}\right\}\left(1 + K[H^+]\right) \tag{4}$$

Or if we replace V by $V_0 + v$, $[OH^-]$ by $K_w/[H^+]$, and $[H^+]$ by 10^{-pH}, we get the following.

$$v^* = \left\{v + \left(10^{-pH} - K_w 10^{pH}\right)\left(\frac{V_0 + v}{C}\right)\right\}\left(1 + 10^{-pH}K\right) \tag{5}$$

An expression similar to eq 4 was derived previously by Ingman and Still (2). Equation 4 or 5 predicts the ep volume v^* at each point along the titration defined using the measurement data pH vs. v and the explicit parameters K_w, K, V_0, and C. The same prediction equation is valid for titration points both before and after the equivalence point and for both weak and strong acidic analytes.

Simplifying the Calculations

If the predictions are to be calculated with a spreadsheet, the complete eq 5 may be used as written. However, if they are done manually, the computational effort is easily reduced whenever good approximations clearly exist.

For example, if the analyte acid is strong or completely dissociated at all points, the term $K\,[H^+]$ disappears. Also, well before the equivalence point, $[OH^-]$ may be negligible relative to $[H^+]$, so $[OH^-]$ need not be calculated. Similarly, well after the equivalence point, $[H^+]$ may not be significant relative to $[OH^-]$.

An Example

The Computer Simulation

The example offered here is a computer simulation of a titration that might well be carried out in an introductory analytical chemistry course. The student uses a 50-mL pipet to transfer an aliquot of a dilute monoprotic acid in a nonprotic electrolyte, say 0.5 M NaCl, into a titration vessel. The titrant is 1.000×10^{-3} M standardized NaOH solution in 0.5 M NaCl delivered from a 10-mL glass buret. The pH meter has a digital display which reads to 0.001 pH. It is calibrated to read pH as $-\log\ [H^+]$ in 0.5 M NaCl.

The titration data are given in the table for the benefit of readers who wish to carry out their own calculations. The same data are plotted as the titration curve in the top panel of Figure 1. This curve has a poorly defined inflection point somewhere in the region of 4 to 6 mL. Because the pH is about 8 at a point roughly half-way to this region, we judge that the acid dissociation pK_a is near 8, so K in eq 1 is near 10^8.

Example of Titration Data

v mL	pH	v mL	pH
0.03	6.212	4.79	8.858
0.09	6.504	4.99	8.926
0.29	6.936	5.21	8.994
0.72	7.367	5.41	9.056
1.06	7.567	5.61	9.118
1.32	7.685	5.85	9.180
1.53	7.776	6.05	9.231
1.76	7.863	6.28	9.283
1.97	7.938	6.47	9.327
2.18	8.009	6.71	9.374
2.38	8.077	6.92	9.414
2.60	8.146	7.15	9.451
2.79	8.208	7.36	9.484
3.01	8.273	7.56	9.514
3.19	8.332	7.79	9.545
3.41	8.398	7.99	9.572
3.60	8.458	8.21	9.599
3.80	8.521	8.44	9.624
3.99	8.584	8.64	9.645
4.18	8.650	8.84	9.666
4.40	8.720	9.07	9.688
4.57	8.784	9.27	9.706

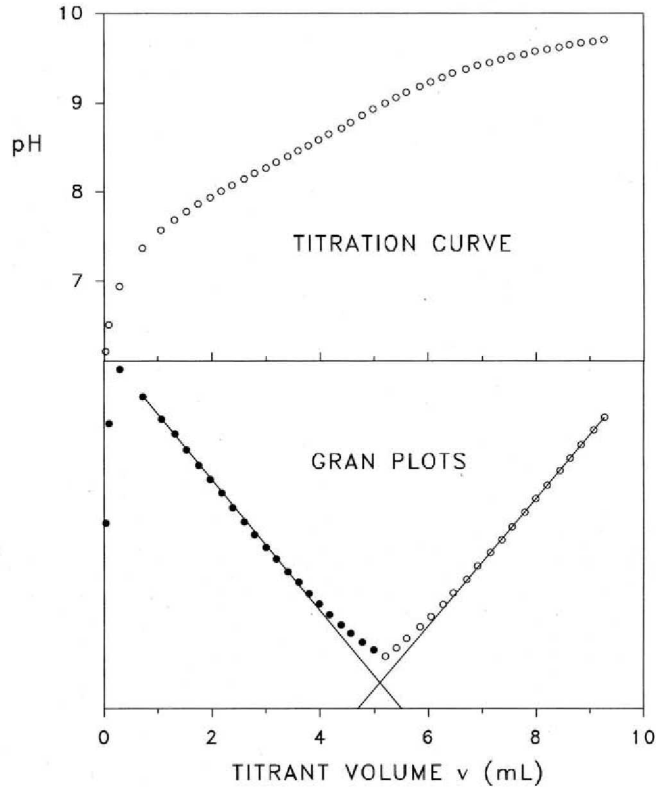

Figure 1. The upper panel is a plot of the titration data used in the example and listed in the table. The lower panel shows Gran plots for these same data.

Construction of Gran Plots

In an attempt to gain a better estimate of the equivalence point, we have constructed two Gran plots (3) in the lower panel of Figure 1. For v values less than about 5 mL, the weak-acid Gran plot is $[H^+]\,v$ vs. v (filled circles). Above 5 mL, the plot is $V/[H^+]$ vs. v (open circles). The ordinates of each Gran plot are scaled so that both plots can be graphed on the same set of axes. The apparent straight segment of each plot is extrapolated to the v axis where the two intersections yield different estimates of the equivalence point: 4.7 and 5.5 mL.

Setting Up the Spreadsheet

At this point we begin to use an electronic spreadsheet to examine the behavior of the titration data through eq 5. We are searching for values of the parameters K (or pK_a) and K_w (or pK_w) that yield the same value for v^* from eq 5 at all 44 titration points.

The spreadsheet is constructed as below.

- Enter the v data in one column.
- Enter the pH data in a second column.
- Put an initial estimate for pK_a, say 8.0, in one cell.
- Put an initial estimate for pK_w, say 14.0, in another cell.
- Enter formulas for v^* as eq 5 in a third column.

These formulas refer to the absolute cell addresses for pK_a and pK_w.

Then we set up an XY graph.

- Use the column of v data for X (the abscissas).
- Use the column of v^* formulas for Y (the ordinates).

If "optimal" values of pK_a and pK_w are entered in their appropriate cells, the v^* values will scatter randomly around a mean value, so the graph called a "v plot" will

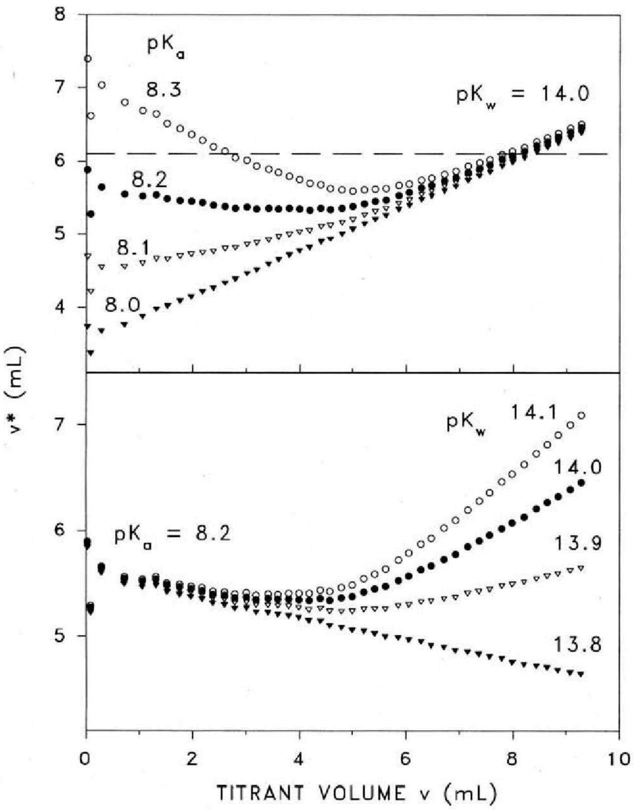

Figure 2. The upper panel shows v plots calculated from eq 5 using fixed pK_w and varying pK_a in steps of 0.1. The lower panel shows v plots calculated using fixed pK_a and varying pK_w. The dashed horizontal line in the upper panel is the mean of points in the v plot with pK_a = 8.3 and pK_w = 14.0.

appear as a horizontal scatter plot. Thus, the v plot serves as a visual monitor of our progress as we search for optimal values of the parameters. When we achieve a horizontal v plot, the column of v^* values serves as a statistical sample from which we can estimate an uncertainty of the ep volume.

The Parameter Search

The parameter search sequence should be done systematically by changing one parameter while holding the other fixed. The top panel in Figure 2 shows four v plots calculated by varying pK_a in steps of 0.1 in the neighborhood of 8 while holding pK_w fixed. Although none of the four are horizontal, this search succeeds in localizing the correct pK_a within the range of 8.1 to 8.2. Here the slopes of the left-hand branches change sign from positve to negative. Thus, this range must include a pK_a that yields a horizontal segment on its v plot.

In the lower panel we show a scan of pK_w in steps of 0.1 while holding pK_a fixed at 8.2. Again we have succeeded in localizing pK_w to within the range of 13.8 to 13.9 because these two right-hand branches must bracket a horizontal segment. Notice that v plots can show the effects due to the separate parameters even though in this example both parameters are simultaneously significant in eq 5 throughout most of the titration.

Monitoring the Progress of the Search

At this point we could continue to close in on the two parameters by scanning in smaller steps and watching the trends in the v plots. However, in order to demonstrate an alternative searching technique we shall use another feature of many electronic spreadsheets. As a preliminary, we calculate a mean value for all the v^* predictions with pK_a= 8.3 and pK_w = 14.0, which is the top v plot in Figure 2. This mean v^* is at 6.1 mL and is shown as the horizontal dashed line in the figure.

Notice that the individual points deviate from this dashed line principally because neither branch is horizontal. However, the points are also somewhat scattered due to random measurement errors.

We can anticipate that as our search brings us closer to the optimal parameter values, three things will happen.

- Both branches of the v plot will become horizontal.
- The entire v plot will come closer to its own v^* mean.
- The scatter of points about that mean will decrease.

Thus, any function that reflects the scatter of points about a mean can serve to monitor the progress of the parameter search. The standard deviation is one such function. If we set up a cell in the spreadsheet that displays the standard deviation of the v^* column, we can watch that standard deviation while changing parameter values.

Using the "What If" Table

Scanning the pK_a as we watch the standard deviation may be systematized by setting up a so-called "sensitivity-analysis" or "what-if" table. This feature will automatically scan ranges of parameter values and display the standard deviation results in the body of a table.

Having already localized the parameters by the v plot searches, we set up such a table with rows of pK_a values from 8.10 to 8.20 in steps of 0.01 and with columns of pK_w values from 13.80 to 13.90, also in steps of 0.01. The minimum standard deviation of v^* appeared in the what-if table in the cell found at row position pK_a = 8.16 and column position pK_w= 13.84.

The v plot for those parameter values is shown on a greatly expanded vertical scale (top panel of Fig. 3). Although this scatter plot is much more horizontal than any in Figure 2, there still appears to be a slight downward trend. Therefore, we refined the search further by revising the what-if table to scan in steps of 0.001 pK around the current values and observed a slightly smaller standard deviation at pK_a = 8.156 and pK_w = 13.840. The corresponding v plot shown in the lower panel of Figure 3 appears to be essentially horizontal, so we terminate the search and accept these two values as optimal.

Results

With this configuration, the mean of the values in the v^* column is 5.055 mL. The standard error estimate for this mean is the sample standard deviation (0.061 mL) divided by the square root of the count (44 points), which turns out to be 0.0092 mL. By multiplying these results by the titrant concentration 1.000×10^{-3} M, we obtain a determination of 5.055 ± 0.009 (2) micromoles for the analyte acid. However, as we will show, a more rigorous determination and statistical uncertainty estimate can be readily calculated within the spreadsheet.

Weighting Factors

By observing the scatter of points about the horizontal in the lower panel of Figure 3, it appears that there is more scatter early and less scatter later in the titration. In fact, two points (0.03, 5.292) and (0.09, 4.743) at the very beginning are not shown because they scatter vertically off-scale. In other words, the data points are heteroscedastic, meaning that their variance is not uniform along the v plot. In such a situation it is proper to calculate a weighted mean and a weighted standard error of the mean so that the less reliable points early in the titration carry less influence in the overall statistics. The weighting factors can

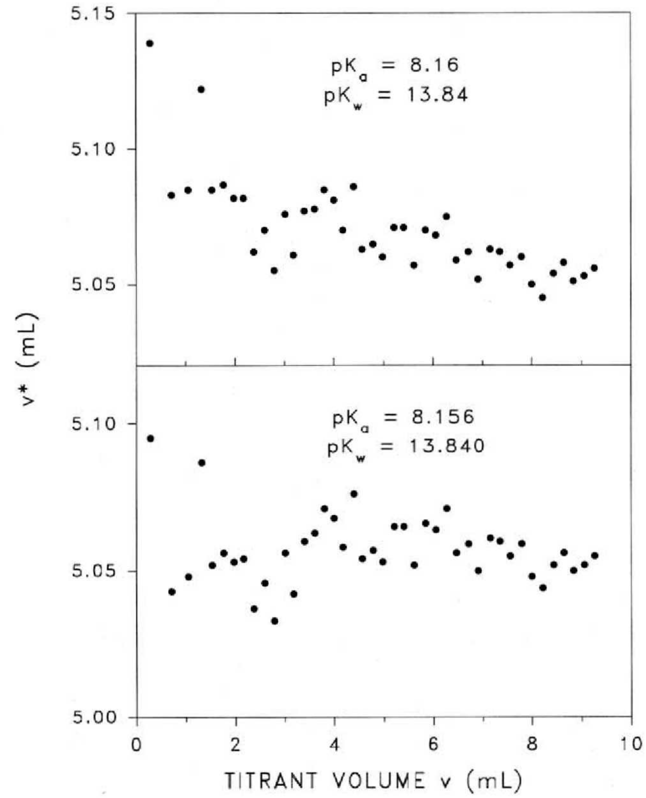

Figure 3. Two *v* plots calculated using parameters as shown. The *v* plot in the lower panel is judged to be horizontal.

be derived from eq 4 or 5, with their calculations included as a separate column in the spreadsheet.

The simulated experimental titration uses a low-noise digital pH meter reading to 0.001 pH. Because all titration solutions have reasonable buffering capacity, the pH measurements can be assumed to have random errors in the order of 0.001 for an average pH of about 8. On the other hand, the 10-mL buret is readable to about 0.01 mL relative to an average reading of about 5 mL. Thus, a reasonable assumption is that the random scatter in the v plot is primarily due to random errors in reading the meniscus level. With this assumption, the variable v in eq 5 is the only random variable on the right-hand side.

If we assume that the standard deviation of errors in reading the meniscus level is a constant but unknown σ, a propagation-of-variance analysis based on eq 4 leads to the variance estimates for v^*.

$$\text{var } v^* = \left(1 + K[\mathrm{H^+}]\right)^2 \left(1 - \frac{[\mathrm{OH^-}]}{C}\right)^2 \sigma^2$$

Weighting factors are inversely proportional to these variances. We calculate a weighted mean ep volume $\bar{v}^*$ from the individual ep volumes v_i^* using the following formula.

$$\bar{v}^* = \frac{\sum w_i v_i^*}{\sum w_i} \tag{6}$$

where the weighting factors are

$$w_i = \left(1 + K[\mathrm{H^+}]\right)^{-2} \left(1 - \frac{[\mathrm{OH^-}]}{C}\right)^{-2} \tag{7}$$

Then the weighted standard error of the mean $s_{\bar{v}^*}$ is estimated from the following equation.

$$s_{\bar{v}^*} = \left(\frac{\sum w_i (v_i^* - \bar{v}^*)^2}{(n-1)}\right)^{1/2} \tag{8}$$

where n is the count of data points (*4*).

These equations are easily programmed into the spreadsheet.

- A column for w_i, using eq 7
- A column for products $w_i v_i^*$
- A cell for $\bar{v}^*$, using eq 6
- A column for formulas $w_i(v_i^* - \bar{v}^*)^2$
- A cell for $s_{\bar{v}^*}$, as the formula of eq 8

Using the optimal v plot obtained above, we find that the weighting factors w_i's vary in a regular way from a low of 0.00013 at the beginning of the titration, to 0.75 in the middle, to a high of 1.1 at the end. Thus, the early points are effectively deemphasized in the summations. The weighted mean $\bar{v}^*$ is 5.057 mL, and its weighted standard error estimate is 0.0065 mL. Multiplying these figures by the titrant concentration yields an analyte determination of 5.057 ± 0.006 (*5*) μmol.

Results and Comments

In reality a student carrying out a laboratory titration would, of course, have no knowledge of the titration parameters, but in our unreal world of computer simulation we do have this information. The 50-mL pipet delivered 49.94 mL of analyte solution having a concentration of 1.020×10^{-4} M monoprotic acid for which the true dissociation constant pK_a was 8.160. The true pK_w was 13.840, and the concentration of NaOH titrant was 1.004×10^{-3} M.

True pH and v values were computed to about 14 decimal places, but the recorded pH measurements were rounded off to the nearest 0.001. Uniformly distributed random errors varying between limits of ±0.01 mL were added to each reading of the meniscus level, and those sums were rounded to the nearest 0.01 mL. All v plot calculations were done with these rounded data. Thus, the true amount of acid analyte was 5.094 μmol.

The discrepancy between this value and the weighted mean of 5.057 μmol (with its random error estimate of 0.0065 mol) is principally due to systematic errors generated by the slightly inaccurate values assumed for the parameters C and V_0. We could have continued our search for optimal parameter values by attempting to adjust C or V_0 or both while monitoring the v^* standard deviation cell. However, there was no incentive to do this because we obtained a horizontal v plot by adjusting the pK_a and pK_w only.

Summary

The method presented here is offered as a means of estimating a statistical error for a titration equivalence point. There are few alternatives. Nonlinear regression analysis is one possibility, and this requires the programming of one of the available algorithms. Indeed, the search for v plot optimal parameters done by spreadsheet manipulations could well be done as a nonlinear regression problem. This would simply involve rearrangement of the model equations from their forms as usually written for nonlinear regression analysis of titrations. The other alternative is based on analysis of the straight segments of Gran plots (*5*), but again this requires a substantial level of computer programming.

A Variation

For those instructors who wish to introduce the subject without using a spreadsheet for parameter searching, we suggest that an example be offered of the titration of a strong acid in constant ionic strength medium. If only data recorded before the equivalence point are used, the v plot equation reduces to

$$v^* = v + 10^{-\mathrm{pH}}\left(\frac{V_0 + v}{C}\right)$$

which involves no adjustable parameters if V_0 and C are accurately known. Thus, the v plot is immediately horizontal, and an unweighted standard deviation can be easily calculated manually using an electronic calculator.

Then at a second level of sophistication, the instructor might assign a single parameter search. This might be based on data recorded after the equivalence point of the strong acid titration for which the v plot equation

$$v^* = v - K_w 10^{\mathrm{pH}}\left(\frac{V_0 + v}{C}\right)$$

involves only K_w as an adjustable parameter.

Alternatively, a single parameter v plot might involve data before the equivalence point of a not-too-weak acid titration for which K is the only adjustable parameter in the v plot equation.

$$v^* = \left\{v + 10^{-\mathrm{pH}}\left(\frac{V_0 + v}{C}\right)\right\}\left(1 + K\,10^{-\mathrm{pH}}\right)$$

In either case, if programmable calculators are not available, the parameter search might be monitored by observing the difference between two well-separated v^* values. This difference reflects the slope of the v plot segment, so the absolute value of the slope will be a minimum when the optimal parameter is found.

Literature Cited

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